📘 1. Unraveling the World of Data and Its Insights

Data is raw information.
Analytics + Statistics convert that raw information into **meaningful insights**, enabling decisions in:

- Business
- Healthcare
- Social media
- Finance
- Education
- Machine Learning

✔ Types of Data

Type of Data	Examples
Numerical	Age, weight, income, marks
Categorical	Gender, city, department
Ordinal	Rating scale (1 to 5), satisfaction level
Time Series	Stock prices, daily sales

✔ Why do we analyze data?

- To **understand** what is happening
- To **explain** why it is happening
- To **predict** what may happen
- To **recommend** what to do next

Population vs Sample

1. What is a Population?

A **population** includes *all* members of a defined group that you want to study or draw conclusions about.

Population Parameters

- **Population Mean** = μ
- **Population Standard Deviation** = σ
- **Population Proportion** = P

2. What is a Sample?

A **sample** is a *small subset* of the population selected for analysis.

Sample Statistics

- **Sample Mean** = $\bar{x}$
- **Sample Standard Deviation** = s
- **Sample Proportion** = p

📘 2. The Basics of Statistics: Descriptive Insights

Descriptive statistics summarize large datasets into simple numbers.

📌 Measures of Central Tendency

5, 7, 7, 9, 10, 12, 13, 15, 15, 18, 20, 40

$$\text{Mean} = \frac{\sum x}{n}$$

Sum of values:

5 + 7 + 7 + 9 + 10 + 12 + 13 + 15 + 15 + 18 + 20 + 40 = **171**

Number of values = 12

$$\text{Mean} = \frac{171}{12} = 14.25$$

⭐ 2. Median (Middle Value)

12 values → even number →

Median = average of 6th and 7th values.

Sorted data:

5, 7, 7, 9, 10, **12**, **13**, 15, 15, 18, 20, 40

$$\text{Median} = \frac{12 + 13}{2} = 12.5$$

⭐ 3. Mode (Most Frequent Value)

Values repeating:

7 → twice

15 → twice

So:

$$\text{Mode} = 7, 15 \quad (\text{bimodal})$$

⭐ 4. Range

$$\text{Range} = \text{Max} - \text{Min} = 40 - 5 = 35$$

⭐ 5. Variance

Step 1: Find the deviation from mean

Mean = 14.25

x	x − mean	(x − mean)²
5	-9.25	85.56
7	-7.25	52.56
7	-7.25	52.56
9	-5.25	27.56
10	-4.25	18.06
12	-2.25	5.06
13	-1.25	1.56
15	0.75	0.56
15	0.75	0.56
18	3.75	14.06
20	5.75	33.06
40	25.75	663.06

Sum of squared deviations = **954.2**

For sample variance (most common):

$$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$$

$$s^2 = \frac{954.2}{11} = 86.74$$

⭐ 6. Standard Deviation

$$s = \sqrt{86.74} = 9.31$$

1 Mean (Average)

$$\text{Mean} = \frac{\sum x_i}{n}$$

Example:

Marks = 10, 20, 30
Mean = (10+20+30)/3 = 20

---

2 Median

Middle value when data is sorted.

- Odd n → middle value
- Even n → average of two middle values

---

3 Mode

Most frequent value
Examples:

- Shoe size most sold = 7
- Website most visited browser = Chrome ✔

📘 3. Measuring Data Spread: Dispersion Insights

Dispersion tells us **how spread out** the data is.

📌 1 Range

$$\text{Range} = \text{Max} - \text{Min}$$

📌 2 Variance

Shows average squared deviation from mean.

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

(population)

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

(sample)

📌 3 Standard Deviation

Square root of variance.
Shows **how far values are from the mean**.

σ=sqrt(σ2)

More SD = more spread
Low SD = stable data

📌 4 Interquartile Range (IQR)

Middle 50% of data.

$$\text{IQR} = Q3 - Q1$$

📘 4. Unlocking Relationships: The Power of Association

Association checks how variables move together.

📌 1 Covariance

$$\text{Cov}(X, Y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- **Positive** → move in same direction
- **Negative** → move in opposite direction
- **0** → no pattern

📌 2 Correlation (Pearson)

Standardized covariance, ranges from −1 to +1.

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_x \sigma_y}$$

r value	Meaning
+1	Perfect positive
-1	Perfect negative
0	No relationship
0.3–0.5	Moderate
> 0.7	Strong

Example:

- Study time vs marks = positive correlation
- Price vs demand = negative correlation

covariance, **Pearson correlation coefficient**, and **skewness** for a worked example so you can see the full math + interpretation.

Note: covariance & correlation need a *pair* of variables. I used the test scores you gave:

```
Y (Scores) = [5, 7, 7, 9, 10, 12, 13, 15, 15, 18, 20, 40]
```

and for demonstration I created a plausible paired variable:

```
X (StudyHours) = [1, 2, 2, 3, 4, 5, 5, 6, 6, 7, 8, 12]
```

If you already have a real paired variable (e.g., StudyHours, Attendance), tell me and I'll re-run with that.

---

1) Quick summary of results (rounded)

- **Mean(Score) = 14.25**
- **Mean(StudyHours) = 5.0833**
- **Sample covariance (Cov(X,Y)) = 27.4318**
- **Pearson correlation r = 0.9629** (strong positive correlation)
- **Skewness (Fisher–Pearson adjusted g_1) = 2.0967** (strong positive skew)
  - Unadjusted skew (simple third moment / s^3) = **1.6016**

---

4) Practical takeaways / interpretation

- **Covariance > 0** and **r ≈ 0.963** → X and Y move together strongly (in our example: more study hours → higher scores).

- **High skewness (≈ 2.10)** shows the distribution of scores is **right-skewed** (long tail to the right). The value **40** is an outlier and inflates mean & variance.
- Because of skew and outlier, the **median (12.5)** is a more robust "center" than the mean (14.25) for describing typical score.
- If you plan to run regressions or parametric tests, consider:
    - transforming data (log) or
    - using robust methods or
    - removing/inspecting outliers before modeling.

Population vs Sample

1. What is a Population?

A **population** includes *all* members of a defined group that you want to study or draw conclusions about.

Examples

- All customers of Zomato in India
- All 8th-grade students in a school
- Every product sold in 2024
- Every employee in a company

Key Points

- Represents the **entire group**
- Usually **large** (sometimes infinite)
- Hard to measure completely
- True parameters come from population (like μ, σ)

Population Parameters

- **Population Mean** = μ
- **Population Standard Deviation** = σ
- **Population Proportion** = P

---

2. What is a Sample?

A **sample** is a *small subset* of the population selected for analysis.

Examples

- 500 Zomato customers selected from all users
- 100 students selected randomly from all 8th graders

- 1000 product records out of 10 lakh sales
- Surveying 200 employees instead of all 5000

Key Points

- Represents a **part** of the population
- Used when studying entire population is impossible
- Should be **random** to reduce bias
- Sample statistics are used to estimate population parameters

Sample Statistics

- **Sample Mean** = $\bar{x}$
- **Sample Standard Deviation** = s
- **Sample Proportion** = p

---

3. Why Do We Use Samples?

Studying the whole population can be:

✔ Expensive
✔ Time-consuming
✔ Impossible (infinite population like website visitors)

Sampling allows:

✔ Faster results
✔ Lower cost
✔ Statistical inference using probability

---

4. Population vs Sample — Comparison Table

Feature	Population	Sample
Definition	Whole group	Subset of population
Size	Very large (N)	Small (n)
Values	Parameters	Statistics

Feature	Population	Sample
Mean	M	x̄
Standard deviation	Σ	s
Analysis	Difficult	Easier
Accuracy	High	Estimate of population
Cost & Time	High	Low

📘 5. Decoding Skewness: Understanding Data Distribution

Skewness measures symmetry.

📌 Formula for Skewness (Fisher's)

$$\text{Skewness} = \frac{3(\bar{x} - \text{median})}{s}$$

Types:

- **Positive skew (Right-skewed):** Mean > Median
- **Negative skew (Left-skewed):** Mean < Median
- **Symmetric:** Mean = Median

📘 6. Unlocking the World of Probability

Probability measures the likelihood of an event.

📌 Formula

$$P(A) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$$

☆ Properties:

- $0 \leq P(A) \leq 1$
- P(Not A) = 1 – P(A)
- P(A and B independent) = P(A) × P(B)
- P(A or B) = P(A) + P(B) – P(A and B)

📘 7. Understanding Uniform Distribution

A uniform distribution has **equal probability** for all outcomes.

✔ Example: Rolling a fair dice

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

Used in simulations, gaming, randomization.

📘 8. The Binary World of Binomial Distribution

Used when data has **two outcomes** → Success or Failure.

📌 Conditions:

✔ Fixed number of trials (n)
✔ Two outcomes (success/failure)
✔ Probability remains constant (p)
✔ Independent trials

📌 Formula

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

📌 Example

Probability of 7 heads in 10 coin tosses.

Normal Distribution (Gaussian Distribution)

The **Normal Distribution** is one of the most important probability distributions in statistics.

It is a **bell-shaped**, **symmetrical**, **continuous** distribution where:

- The **mean = median = mode**
- Most data lies close to the **center**
- Data gradually decreases toward both extremes

📌 Properties of Normal Distribution

1. **Bell-shaped & symmetrical**
   Left and right sides mirror each other.
2. **Mean = Median = Mode**
   All three located at the center.
3. **Total area under the curve = 1 (100%)**
4. **68–95–99.7 Rule (Empirical Rule)**
   - **68%** of data lies within **±1 standard deviation**
   - **95%** within **±2 standard deviations**
   - **99.7%** within **±3 standard deviations**

The Z-score *standardizes* any normally distributed data

If a dataset follows (or approximately follows) a **normal distribution**, you can convert every value into a **Z-score** using:

After this transformation, the entire dataset becomes a **Standard Normal Distribution** with:

- Mean = 0
- Standard deviation = 1

- $$Z = \frac{X - \mu}{\sigma}$$

This new distribution is called the **Z-distribution**.

📘 9. The Z-Score Journey (Standard Score)

Z-score tells how many **standard deviations** a value is from the mean.

📌 Formula-

$$Z = \frac{x - \mu}{\sigma}$$

📌 Interpretation

Z-Score	Meaning
0	exactly on mean
+1	1 SD above mean
-1	1 SD below mean
> ±3	Outlier

Used in:

- Outlier detection
- Standard normal distribution
- Hypothesis testing

📘 10. The T-Score (Student's t-statistic)

Used when:
✔ Sample is small (n < 30)
✔ Population SD is **unknown**

📌 Formula

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Uses

- Confidence intervals
- Hypothesis testing
- A/B testing
- Small sample research

---

when to apply z-test and when p-value?

ChatGPT said:

```
import numpy as np from statsmodels.stats.weightstats import ztest

# sample data

sample = np.array([45, 52, 47, 49, 51, 46, 48, 50])

 # perform one-sample z-test

 z_stat, p_value = ztest(sample, value=50)

print("Z-statistic:", z_stat)

print("p-value:", p_value)
```

Here's the **clear, simple rule**:

✅ Z-test vs p-value

1 Z-test

A **Z-test** is a **type of hypothesis test** you use when:

✔ Use Z-test when:

- **Population standard deviation (σ) is known**
- Sample size is **large (n ≥ 30)** (by CLT)
- Data is **approximately normally distributed**

Examples:

- Comparing **sample mean vs population mean**
- Comparing **two proportions** (large samples)

Output of a Z-test:

- **Z-statistic**
- **p-value**

🟦 → **Z-test is the method.**
🟦 → **Z-statistic is the test statistic.**

---

2 p-value - the p-value = probability of getting a test statistic as extreme or more extreme than your sample's value IF the null hypothesis were true.

A **p-value is NOT a separate test**.
It is the **result of any hypothesis test** (Z-test, t-test, χ^2-test, F-test, etc.).

✔ Use p-value to:

- Decide whether to **accept or reject the null hypothesis**

Decision rule:

- If **p-value < α** (usually 0.05) → **Reject H_0**
- If **p-value ≥ α** → **Fail to reject H_0** (accept H_0)

🎯 So what is the difference?

Z-test	p-value
A hypothesis testing method	A result/decision measure
Produces a Z-score	Produced by the Z-score
Used only when sample is large and σ known	Used for ANY test

🎓 Intuition

- You **first choose a test** (Z-test).
- That test gives you a **Z-value**.
- That **Z-value gives a p-value**.
- **You use the p-value** to decide accept/reject.

📌 Example

Suppose Z = 2.98

- Convert using Z-table → **p = 0.003**
- $p < 0.05$ → **Reject H_0**

Even if Z is "within 3 SD",
the p-value is the true decision maker, NOT just the SD rule.

how Z-value gives p-value?

ChatGPT said:

Here's the **simple, clear explanation** of how a **Z-value gives a p-value** 👇

✅ How Z-value gives p-value

A **Z-value** tells you **how many standard deviations** your sample result is from the mean **under the null hypothesis**.

Once you have the Z-value, you convert it into a **probability** using the **standard normal distribution** curve.

This probability = **p-value**.

☆ Step-by-step (super simple)

1 You compute Z

$$Z = \frac{\text{sample statistic} - \text{population mean}}{\text{standard error}}$$

Example:
Z = **2.98**

---

2 Plot this Z on the standard normal curve

The standard normal curve has:

- Mean = 0
- SD = 1

So Z = 2.98 means the result is **2.98 SD away from the centre**.

---

3 Look up the area in the tail

For a **right-tail test**:
p-value = area **to the right** of Z = 2.98

This is found using:

- Z-table
- Excel
- Python
- Calculator

The area in the tail beyond **Z = 2.98** is:

p=0.0014 (approx)

🎯 Why is this the p-value?

Because the **p-value = probability of getting a test statistic as extreme or more extreme than your sample's value IF the null hypothesis were true**.

Z = 2.98 is far in the tail → rare → low probability → **reject H_0**.

🔢 Formulas to convert Z → p-value

Right-tailed test

$$p = 1 - \Phi(Z)$$

Left-tailed test

$$p = \Phi(Z)$$

Two-tailed test

$$p = 2 \times (1 - \Phi(|Z|))$$

Where Φ(z) = cumulative distribution function (area under the curve).

🧪 Example in Excel

If Z = 2.98:

Right-tailed

```
=1 - NORM.S.DIST(2.98, TRUE)
```

Two-tailed

```
=2 * (1 - NORM.S.DIST(ABS(2.98), TRUE))
```

📌 Intuition

- Z = 0 → p = 0.5 (not extreme)
- Z = 1 → p = 0.158 (mild)
- Z = 2 → p = 0.0228 (rare)
- Z = 3 → p = 0.0013 (very rare)

So **higher |Z| → lower p-value → stronger evidence against H_0**.

#----------

what is pandas in ml and why we use it? how numpy is different then pandas? why we import numpy when importing pandas?

ChatGPT said:

Here is a **simple and clear explanation** suitable for interviews, teaching, and practical ML use 👇

✅ What is Pandas in Machine Learning?

Pandas is a Python library used for data manipulation and analysis.
In Machine Learning, **Pandas is used to load, clean, transform, explore, and prepare datasets before feeding them into ML models.**

Think of Pandas as **Excel inside Python**, but much more powerful.

✔ Pandas is used for:

- Reading data (CSV, Excel, SQL, JSON)
- Cleaning data (remove nulls, duplicates, incorrect values)
- Feature engineering
- Filtering & slicing rows/columns
- Grouping & aggregation
- Joining/merging datasets
- Handling time-series data

📌 **Machine learning models (Sklearn, XGBoost, etc.) expect clean numeric data → Pandas prepares that data.**

✅ How is NumPy different from Pandas

Feature	NumPy	Pandas
Primary structure	`ndarray` (numeric array)	`DataFrame` & `Series` (table-like)
Best for	Numerical computation	Data manipulation & cleaning
Data types	Mostly numeric	Numeric + strings + dates
Indexing	Only position-based	Label + position-based
Handling missing values	Not good	Very good (`NaN` handling)
Operations	Fast math operations	Higher-level operations (groupby, merge, join)

Feature	NumPy	Pandas
Use case	ML model calculations, vectorization	Preparing dataset for ML

NumPy = low-level, fast mathematical engine
👉 **Pandas = high-level data manipulation tool built on NumPy**

Why do we import NumPy when using Pandas?

Because:

✔ Pandas is built on top of NumPy

- Pandas **DataFrame internally uses NumPy arrays** to store data.
- All numeric operations inside Pandas are powered by **NumPy**.

Example:
Pandas column → internally it's a NumPy array:

```
df['age'].values  # returns numpy array
```

Many Pandas functions rely on NumPy features

- Vectorization
- Broadcasting
- Mathematical operations
- NaN handling

So even if you only work with Pandas, **NumPy is still required under the hood**.

- **Pandas is a data analysis library used for cleaning, transforming, and preparing data in ML.**

- **NumPy is a numerical computation library.**

- **Pandas is built on top of NumPy; therefore, NumPy is required whenever you use Pandas.**

#----------